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ENGINEERING
TEXAS A&M UNIVERSITY

Team 64: Sound Localization Robot 4 Bi-Weekly Update 4

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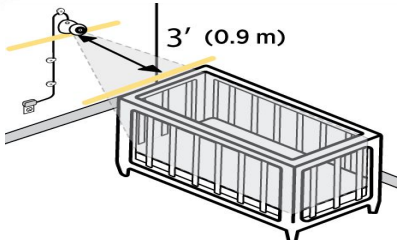
TA: Roman Venegas

Project Summary

Problem Statement:

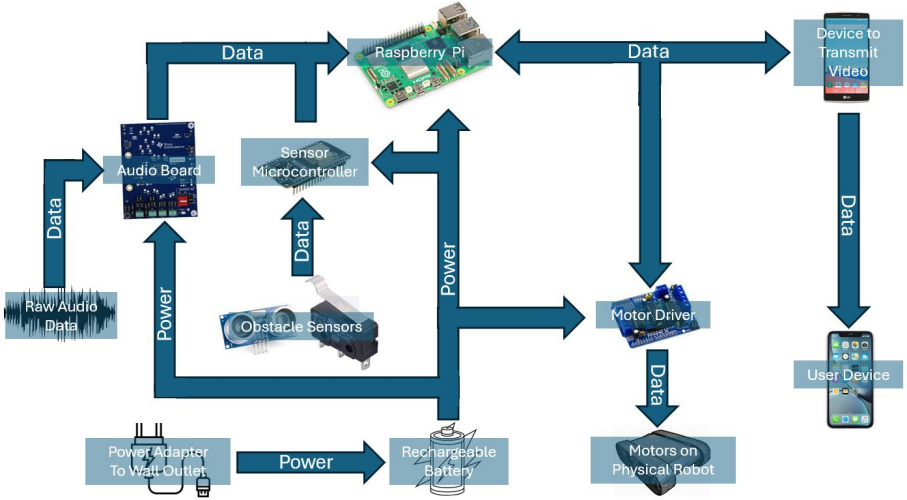
- Current baby monitors are stationary and cover only a limited area
- Some offer camera movement, but still require caretaker control



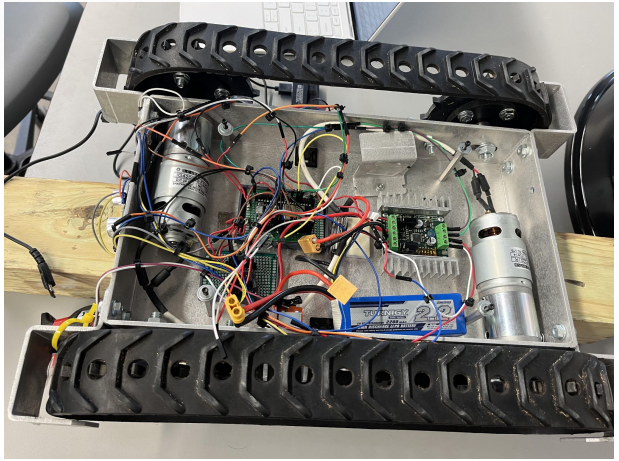
Existing monitors only cover a narrow area

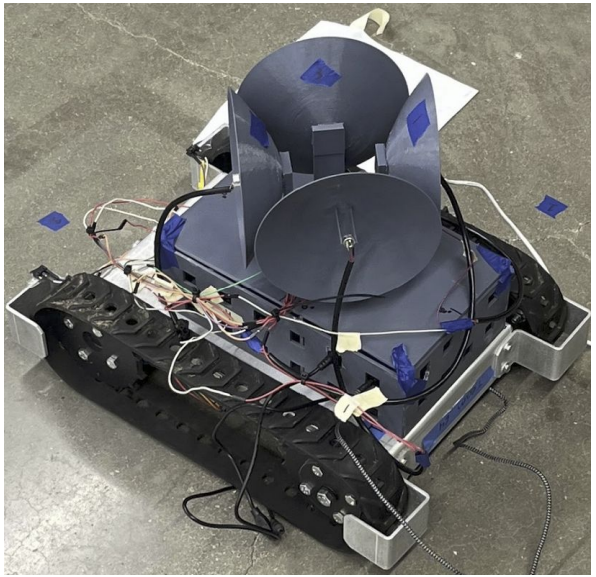
Our Solution:

- A robot that autonomously navigates towards a baby's cry then using sound localization, avoiding obstacles along the way
- Starts a video call to show the baby, giving caretakers peace of mind

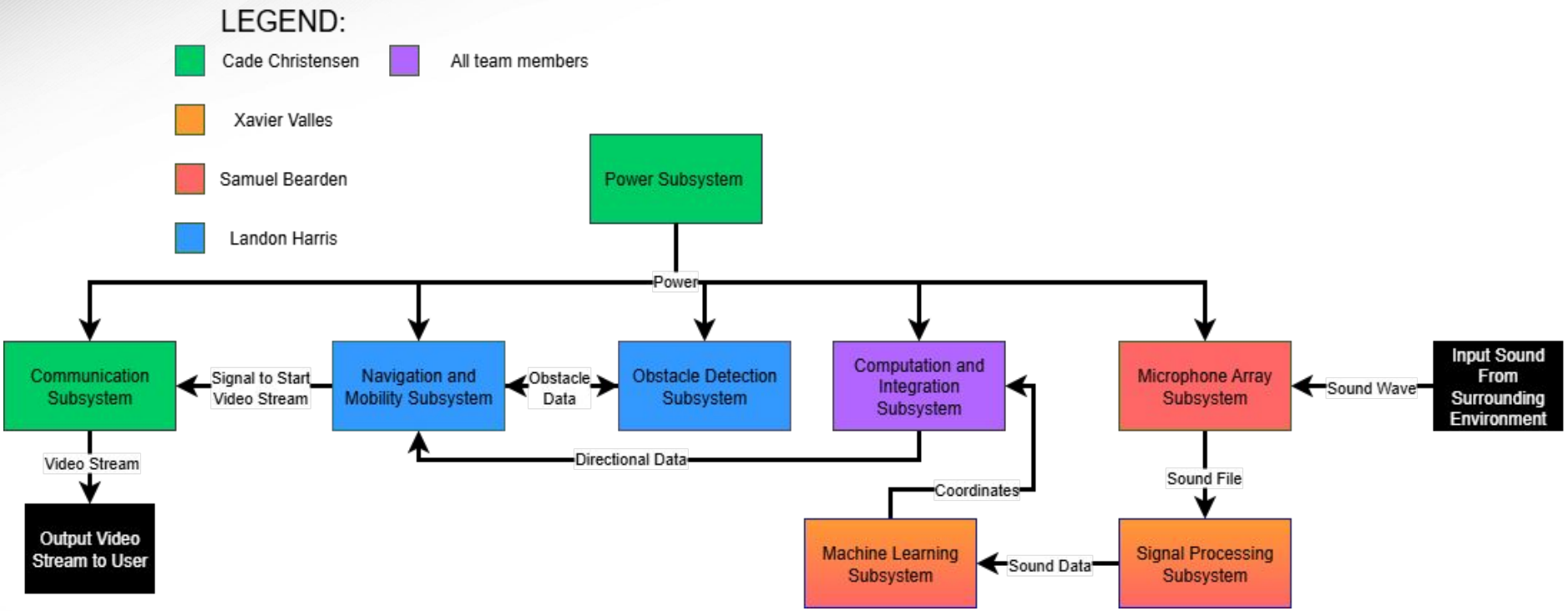


Integrated System Diagram





System Overview



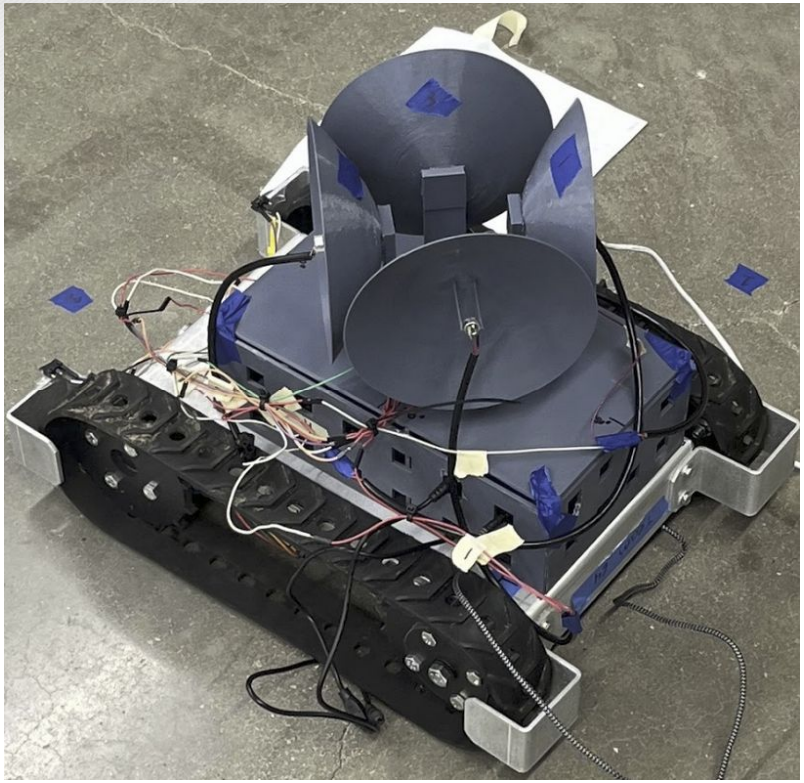
Project Timeline

Subsystem Designs and Testing (completed 1/26)	Debug Subsystem Integrations (to complete by 2/9)	Full Subsystem Integration (to complete by 2/23)	Test Autonomous Robot (to complete by 3/23)	Validations (to complete by 3/30)	Prepare Demo (to complete by 4/6)	Report (to complete by 4/20)

Power and Communication

Cade Christensen

Accomplishments since last update 18 hrs of effort	Ongoing progress/problems and plans until the next presentation
Printed additional housing for the TI board and Raspberry Pi and mounted on the robot Permanently mounted the parabolic dishes on the robot Removing debugging visuals from the app to make it look more finished	Validate the dishes operate well at 10 feet and adjust orientation as needed Verify subsystems work when doing a full system demo Epoxy the housing as the 3D print was warped on printing



Sound Characterization & Noise Filtering

Xavier Valles

Accomplishments since last update 40 hrs of effort	Ongoing progress/problems and plans until the next presentation
Dual channel voting works correctly during live audio detection Raspberry Pi saves TI board configuration at startup and change register values as needed Robot starts autonomously after a single command	Verify Sound Characterization to detect cries past 10 feet by increasing microphone SNR Verify system is effective up to 65 dB under typical household sounds Verify system is maintains accuracy above 90% during edge cases

Sound Characterization & Noise Filtering

Xavier Valles

```
bthz@raspberrypi:/ $ arecord -l
**** List of CAPTURE Hardware Devices ****
card 2: T20 [TI USB Audio 2.0], device 0: USB Audio [USB Audio]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
bthz@raspberrypi:/ $ sudo i2cdetect -y 1
     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:                         -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: 40 -- -- -- -- -- -- -- UU -- -- -- -- -- -- --
50: -- -- -- -- 54 -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- 65 -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
bthz@raspberrypi:/ $
```

Sound Localization Algorithm & Microphone Array

Samuel Bearden

Accomplishments since last update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Fully Built Microphone Array • Fully Integrated Sound Localization Algorithm and Microphone Array with the rest of the System • Retrained and Tested the Sound Localization Algorithm	• Need to Retrain Sound Localization Model Again due to Change of Microphone Gain Settings • Once Trained, a Full System Test Can be Done

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Sound Localization Algorithm & Microphone Array

Samuel Bearden

Navigation/Mobility, & Obstacle Avoidance

Landon Harris

Accomplishments since last update 24 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Tested full robot system, but inconsistencies occurred with the turning angles and distance traveled • Developed encoder testing scripts for distance and rotation measurements • Mounted and tested wheel encoder system • Integrated encoder feedback into the navigation system	• Test and debug full system with encoder feedback • Test and debug the obstacle avoidance with sound localization implementation

Navigation/Mobility, & Obstacle Avoidance

Landon Harris

Trial	Encoder Counts	Distance Traveled (Cm)
1	24	42
2	24	41
3	24	39

Trial	Encoder Counts	Measured Angle (Cm)
1	16	90
2	16	87
3	16	92

Execution Plan

	1/12/2026	1/19/2026	1/26/2026	2/2/2026	2/9/2026	2/16/2026	2/23/2026	3/2/2026	3/9/2026	3/16/2026	3/23/2026	3/30/2026	4/6/2026	4/13/2026	4/20/2026	4/27/2026	Date
Status Update 1																	1/26/2026
Software Integration												Not Started					2/9/2026
Status Update 2												In Progress					2/9/2026
Sound Localization/Cry Detection Integration												Completed					2/20/2026
Obstacle Avoidance/Sound Localization Integration												Behind Schedule					2/20/2026
Cry Detection/Mobile App Integration																	2/20/2026
Design Blitz Participation																	2/20/2026
Status Update 3																	2/23/2026
Obstacle Avoidance/Mobile App Integration																	2/23/2026
Power System Integration with Motors and Boards																	2/23/2026
Status Update 4																	3/16/2026
System Test																	3/23/2026
Physical Design Clean Up																	3/27/2026
Status Update 5																	3/30/2026
System Validation																	3/30/2026
Prepare Demo																	4/6/2026
Trial Demo with TA																	4/9/2026
Final Design Presentation																	4/13/2026
Trial Demo with other TA																	4/16/2026
Final System Demo																	4/21/2026
Showcase																	4/24/2026
Final Report																	4/27/2026

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Validation Plan

Paragraph #	Test Name	Success Criteria	Methodlogy	Status	Responsible Engineer(s)
3.2.1.1.	Cry Detection Accuracy	**The system correctly identifies an infant cry at least 90% of the time under typical household noise conditions. False positives remain below 10%**	Play recorded infant cry samples at varying volumes in a typical household environment with background noise. Record detection rate and false positive rate from ambient sounds.	Success	Xavier Valles
3.2.1.2.	Sound Localization Accuracy	**The system correctly localizes the sound source (room or zone) at least 90% of the time in typical indoor conditions.**	The system will be tested at different angles around the robot. A successfull test is one with a 90% accuracy in distance and direction	Success	Samuel Bearden
3.2.1.3.	Sound Localization Range	**The system correctly localizes the sound source (room or zone) from at least 10 feet away from the source**	The system will be tested at different distances up to10 feet of the microphone array. A successfull test is one with a 90% accuracy in distance and direction	Success	Samuel Bearden
3.2.1.4.	Navigation Response Time	**The system autonomously navigates to the identified room within 3 minutes of cry detection, accounting for obstacles and typical indoor conditions**	Place system in starting location. Trigger cry sound in target room. Use stopwatch to measure time from cry detection to system arrival in target room. Repeat across various room configurations	Untested	Landon Harris/Samuel Bearden
3.2.1.5	Obstacle Detection Accuracy	**The system detects obstacles in its path with at least 95% accuracy within a detection range of up to 2 meters.**	Place various obstacles at measured distances from 0.2m to 2m in system's path. Record detection events vs actual obstacles. Calculate detection accuracy percentage.	Success	Landon Harris
3.2.2.1	Mass	**The mass of the system should be ≤ 15 pounds.**	Weigh the complete assembled system including all components, batteries, and sensors using a calibrated digital scale. Record mass in both pounds and kilograms. Verify total mass does not exceed 15 lbs	Untested	Full Team
3.2.2.2	Volume Envelope	**The volume envelope of the system should be less than or equal to 293 millimeters in height, 314 millimeters in width, and 458 millimeters in length.**	Measure the complete assembled system dimensions using calipers and measuring tape. Measure maximum height, width, and length of the system including all protruding components. Verify dimensions are within specified envelope: H ≤ 293 mm, W ≤ 314 mm, L ≤ 458 mm.	Untested	Full Team
3.2.2.3	Mounting	**The system should require no permanent installation and be operable on standard household flooring.**	Place system on various standard household flooring types. Verify system operates autonomously without any mounting hardware, adhesives, or permanent fixtures. Test navigation and stability on each surface type. Confirm system can be moved and relocated by user without tools or installation procedures.	Untested	Full Team
3.2.3.1.1	Power Consumption	**The system operates without exceeding 60 W peak power under normal operation.**	Connect system to power meter/analyzer during normal operation cycle including idle, listening, navigating, and obstacle avoidance modes. Record peak power draw. Verify stays below 60W threshold.	Untested	Cade Christensen
3.2.3.1.2	Input Voltage Level	**The system operates correctly with input voltage between 22–29 VDC.**	Use variable DC power supply to test system functionality at 22V, 25.5V (nominal), and 29V. Verify all subsystems function correctly at each voltage level. Monitor for any degradation or errors.	Success	Cade Christensen
3.2.3.2.2	Diagnostic Output	**The system correctly diagnoses errors and reports them accordingly**	The system reports errors correctly when they arise. The different errors depend of different tests, but success is determined by the ability to recognize and report alll errors	Untested	Samuel Bearden
3.2.4.1	Operating Temperature Range	**The system operates reliably within typical household temperature ranges of (68°F to 72°F)**	Place system in temperature-controlled environment. Test full functionality at 68°F, 70°F, and 72°F. Verify cry detection, localization, and navigation perform within specifications at all temperatures.	Untested	Cade Christensen
3.2.4.2	Noise Tolerance	**The cry detection system remains effective in background household noise levels up to 65 dB**	Generate controlled background noise at measured levels (50dB, 55dB, 60dB, 65dB) using typical household sounds. Play infant cry samples at each noise level. Verify detection accuracy remains ≥90%.	In Progress	Xavier Valles
N/A	Power System	**The robot has a consistent source of power to each component with not throttling of current or systems turning off (5V5A for Raspberry Pi 5, 5V .1A for the TI board, 3.3V to the ESP32, and 22V to the motors)**	Connect a calibrated power meter/analyzer to each subsystem rail. Verify 5V/5A to Raspberry Pi 5, 5V/0.1A to the TI PCM6260 board, 3.3V to the ESP32, and 22V to the motors. Run the full system through idle, listening, localization, and navigation modes. Confirm no throttling, voltage drops, or unexpected shutdowns occur during any operational phase.	Untested	Full Team
N/A	Mobile App System	**The mobile app triggers the stream, mobile notification, and email when the robot reaches its destination**	With the full system running, trigger a baby cry event at the source location. Verify the mobile app automatically initiates the live video stream. Confirm a push notification is sent to the caregiver's device. Verify an email alert is sent upon robot arrival at the source. Repeat three times and confirm all three notification types are triggered correctly each time.	Untested	Full Team
N/A	Physical Design System	**Parabolic dishes increase the range of the sound detection to 10 feet Each board is secured to the robot in a 3D printed case so there is no contact to the metal**	Attach parabolic dish enclosures to the microphone array and play a cry sample at 10-foot range. Verify detection accuracy meets the ≥90% threshold with dishes installed. Inspect all 3D-printed board enclosures to confirm no PCB or component contact with the robot metal chassis. Perform a visual inspection and continuity/short-circuit check on each enclosed board before and after a full navigation run.	Untested	Full Team
N/A	Sound Localization System	**The sound localization algorithm correctly recieves the .wav file from the characterization model. The robot accurately pinpoints the sound and sends the information to the navigation subsystem**	Run the cry detection model with a known cry audio file. Verify the characterization model outputs a correctly formatted .wav file and passes it to the localization algorithm. Confirm the localization model produces a valid direction and distance estimate. Verify the output is correctly received by the navigation subsystem and that the robot begins moving toward the estimated source. Test at multiple source positions around the robot.	Untested	Full Team
N/A	Sound Characterization System	**The characterization algorithm correctly identifies whether or not the sound is a baby or not. The characterization algoritm sends the sound file to the localization algorithm**	Play a set of 20 audio clips (10 baby cry, 10 non-cry household sounds) through the PCM6260 board and microphone array. Verify the characterization algorithm classifies each clip correctly with ≥90% accuracy. For each positive cry detection, confirm the algorithm packages and transmits the audio file to the sound localization algorithm within the real-time processing window. Log all classification results and pipeline handoff events for review.	In Progress	Full Team
N/A	Obstacle Detection System	**The robot recieves the location from the localization algorithm. The robot successfully moves to the source of the baby cry and triggers the mobile app to start. The robot successfully navigates obstacles placed in its way.**	Place the robot in a test environment with at least 3 obstacles positioned between the robot and the cry source. Trigger a cry detection event and confirm the robot receives the source location from the localization algorithm. Verify the robot navigates to the source while successfully avoiding all obstacles. Upon arrival, confirm the mobile app notification and stream are triggered. Repeat across two different obstacle configurations and verify consistent behavior.	Untested	Full Team

Thank You For Listening!
Questions?